

Introduction to the Project

The Hotel Management System (HMS) is a basic program that assists in overseeing hotel tasks such as guest registration, room reservations, and invoicing. Python is utilized for programming, while SQLite is employed for the database.

Main Characteristics:

Register customers: Record customer information such as name, address, and contact details.

Booking rooms and determining room fees will be done by Room Booking, taking into account the room category and length of stay.

Bill Management: Keep track of and compute costs for various services such as dining expenses, gaming transactions, and fashion retail buys.

Total Bill Calculation: Determine the total amount due by adding up all charges for a customer.

Search for customers and access their details as well as their booking history.

Structure of a database:

Customer Details (C_DETAILS): Records details of customers.

Room Rent (ROOM_RENT): Maintains records of room reservations and costs.

Bills for services (RESTAURANT_BILL, GAMING_BILL, FASHION_BILL): Fees for various types of services provided by the store.

TOTAL_BILL is where the complete payment for every client is recorded.

Source code

```
import sqlite3

# GLOBAL VARIABLES DECLARATION
connection = None
cursor = None
userName = ""
password = ""
roomRent = 0
restaurantBill = 0
gamingBill = 0
fashionBill = 0
bookingCharge = 500 # Fixed booking charge
totalAmount = 0
cid = ""
new_customer_added = False # Flag to track if a new customer is added

# MODULE TO CHECK SQLITE CONNECTION
def SQLiteconnectionCheck():
    global connection
    try:
        connection = sqlite3.connect("HMS.db") # SQLite will create the file if it doesn't
        exist
        cursor = connection.cursor()
        cursor.execute('''
            CREATE TABLE IF NOT EXISTS C_DETAILS (
                CID TEXT PRIMARY KEY,
                C_NAME TEXT,
                C_ADDRESS TEXT,
                C_AGE TEXT,
                C_COUNTRY TEXT,
                P_NO TEXT,
                C_EMAIL TEXT
            )
        ''')
        print("\nCONGRATULATIONS! Your SQLite connection has been established!")
        cursor.close()
    except sqlite3.Error as err:
        print(f"\nError establishing SQLite connection: {err}")
        return None
    return connection
```

MODULE TO ESTABLISH SQLITE CONNECTION

```
def SQLiteconnection():
    global connection
    try:
        connection = sqlite3.connect("HMS.db") # Connecting to SQLite database file
        if connection:
            return connection
    except sqlite3.Error as err:
        print(f"\nError establishing SQLite connection: {err}")
        connection.close()
```

User Entry Module

```
def userEntry():
    global cid, new_customer_added
    if connection:
        cursor = connection.cursor()
        while True:
            print("\n1. Add New Customer")
            print("9. Exit")
            choice = input("\nEnter your choice: ")

            if choice == '1':
                cid = input("Enter Customer Identification Number: ")
                if not cid.strip():
                    print("Customer ID is required. Please enter a valid Customer ID.")
                else:
                    # Check if the entered Customer ID already exists
                    cursor.execute("SELECT cid FROM C_DETAILS WHERE cid = ?", (cid,))
                    existing_cid = cursor.fetchone()
                    if existing_cid:
                        print("Customer ID already exists. Please enter a unique Customer ID.")
                    else:
                        break
            elif choice == '9':
                print("Exiting program.")
                return
            else:
                print("Invalid choice! Please select 1 to enter Customer ID or 9 to exit.")

        name = input("Enter Customer Name: ")
        address = input("Enter Customer Address: ")
        age = input("Enter Customer Age: ")
        nationality = input("Enter Customer Country: ")
        phoneno = input("Enter Customer Contact Number: ")
```

```

email = input("Enter Customer Email: ")

# Insert data into the table
sql = "INSERT INTO C_DETAILS
(CID,C_NAME,C_ADDRESS,C_AGE,C_COUNTRY,P_NO,C_EMAIL) VALUES (?, ?, ?, ?, ?, ?, ?, ?)"

values = (cid, name, address, age, nationality, phoneno, email)
cursor.execute(sql, values)
connection.commit()
print("\nNew Customer entered in the system successfully!")
new_customer_added = True # Set the flag to True after adding a new customer
cursor.close()
else:
    print("\nError establishing SQLite connection!")

# BOOKING MODULE
def bookingRecord():
    global cid, roomRent, new_customer_added

    if new_customer_added:
        print("\nNew customer added, skipping Customer ID prompt.")
    else:
        print("Enter customer details to make a booking.")
        customer_detail = input("Enter Customer ID or Name: ")

        # Use the entered detail to search for the customer
        cursor = connection.cursor()
        cursor.execute("SELECT * FROM C_DETAILS WHERE CID = ? OR C_NAME = ?",
(customer_detail, customer_detail))
        customer = cursor.fetchone()

        if customer:
            cid = customer[0] # Accessing by index instead of key
        else:
            print("Customer not found!")
            return

    if connection:
        cursor = connection.cursor()
        cursor.execute('''
            CREATE TABLE IF NOT EXISTS ROOM_BOOKINGS (
                CID TEXT,
                ROOM_CHOICE INT,
                START_DATE TEXT,
                END_DATE TEXT,

```

```
        TOTAL_PRICE INT,  
        FOREIGN KEY(CID) REFERENCES C_DETAILS(CID)  
    )  
'''
```

```
print("\n##### We have the following rooms for you #####")  
print("1. Ultra Royal > 10000 Rs. per day")  
print("2. Royal > 5000 Rs. per day")  
print("3. Elite > 3500 Rs. per day")  
print("4. Budget > 2500 Rs. per day")  
print("5. Exit")
```

```
room_choice = int(input("Enter your choice: "))
```

```
if room_choice == 5:  
    print("Exiting the process. No room booked.")  
    return
```

```
start_date = input("Enter Start Date (YYYY/MM/DD): ")  
end_date = input("Enter End Date (YYYY/MM/DD): ")
```

```
from datetime import datetime
```

```
try:  
    start_date_obj = datetime.strptime(start_date, '%Y/%m/%d')  
    end_date_obj = datetime.strptime(end_date, '%Y/%m/%d')  
    days_of_stay = (end_date_obj - start_date_obj).days  
  
    if days_of_stay <= 0:  
        print("Invalid date range. End date must be after start date.")  
        return  
except ValueError:  
    print("Invalid date format. Please enter dates in YYYY/MM/DD format.")  
    return
```

```
if room_choice == 1:  
    roomRent = days_of_stay * 10000  
elif room_choice == 2:  
    roomRent = days_of_stay * 5000  
elif room_choice == 3:  
    roomRent = days_of_stay * 3500  
elif room_choice == 4:  
    roomRent = days_of_stay * 2500  
else:  
    print("Invalid choice! Exiting the process.")
```

```
return
```

```
sql = "INSERT INTO ROOM_BOOKINGS (CID, ROOM_CHOICE, START_DATE,
END_DATE, TOTAL_PRICE) VALUES (?, ?, ?, ?, ?)"
values = (cid, room_choice, start_date, end_date, roomRent)
cursor.execute(sql, values)
connection.commit()
print("Thank you, your room has been booked from", start_date, "to", end_date, "for",
days_of_stay, "days.")
print("Your total booking price is: Rs.", roomRent)
cursor.close()
else:
    print("\nError establishing SQLite connection!")
```

ROOM RENT MODULE

```
def roomRent():
    global cid, roomRent, new_customer_added

    if new_customer_added:
        print("\nNew customer added, skipping Customer ID prompt.")
    else:
        print("Enter customer details to calculate room rent.")
        customer_detail = input("Enter Customer ID or Name: ")

        # Use the entered detail to search for the customer
        cursor = connection.cursor()
        cursor.execute("SELECT * FROM C_DETAILS WHERE CID = ? OR C_NAME = ?",
(customer_detail, customer_detail))
        customer = cursor.fetchone()

        if customer:
            cid = customer[0] # Accessing by index instead of key
        else:
            print("Customer not found!")
            return

    if connection:
        cursor = connection.cursor()
        cursor.execute('''
            CREATE TABLE IF NOT EXISTS ROOM_RENT (
                CID TEXT,
                ROOM_CHOICE INT,
                NO_OF_DAYS INT,
                ROOM_NO INT,
                ROOM_RENT INT,
```

```
FOREIGN KEY(CID) REFERENCES C_DETAILS(CID)
```

```
)  
'''
```

```
print("\n##### We have the following rooms for you #####")  
print("1. Ultra Royal > 10000 Rs.")  
print("2. Royal > 5000 Rs.")  
print("3. Elite > 3500 Rs.")  
print("4. Budget > 2500 Rs.")  
print("5. Exit")
```

```
room_choice = int(input("Enter your choice: "))
```

```
if room_choice == 5:  
    print("Exiting the process. No room rent calculated.")  
    return
```

```
room_no = int(input("Enter Customer Room No: "))  
no_of_days = int(input("Enter No. of Days: "))
```

```
if room_choice == 1:  
    roomRent = no_of_days * 10000  
elif room_choice == 2:  
    roomRent = no_of_days * 5000  
elif room_choice == 3:  
    roomRent = no_of_days * 3500  
elif room_choice == 4:  
    roomRent = no_of_days * 2500  
else:  
    print("Invalid choice! Exiting the process.")  
    return
```

```
sql = "INSERT INTO ROOM_RENT (CID, ROOM_CHOICE, NO_OF_DAYS,  
ROOM_NO, ROOM_RENT) VALUES (?, ?, ?, ?, ?)"  
values = (cid, room_choice, no_of_days, room_no, roomRent)  
cursor.execute(sql, values)  
connection.commit()  
print("Thank you, your room has been booked for:", no_of_days, "days")  
print("Your total room rent is: Rs.", roomRent)  
cursor.close()  
else:  
    print("\nError establishing SQLite connection!")
```

```
# RESTAURANT MODULE  
def restaurant():
```

```

global cid, restaurantBill, new_customer_added

if new_customer_added:
    print("\nNew customer added, skipping Customer ID prompt.")
else:
    print("Enter customer details for restaurant bill.")
    customer_detail = input("Enter Customer ID or Name: ")

    # Use the entered detail to search for the customer
    cursor = connection.cursor()
    cursor.execute("SELECT * FROM C_DETAILS WHERE CID = ? OR C_NAME = ?",
(customer_detail, customer_detail))
    customer = cursor.fetchone()

    if customer:
        cid = customer[0] # Accessing by index instead of key
    else:
        print("Customer not found!")
        return

if connection:
    cursor = connection.cursor()
    cursor.execute('''
        CREATE TABLE IF NOT EXISTS RESTAURANT (
            CID TEXT,
            CUISINE TEXT,
            QUANTITY INT,
            BILL INT,
            FOREIGN KEY(CID) REFERENCES C_DETAILS(CID)
        )
    ''')

    print("1. Vegetarian Combo > 300 Rs.")
    print("2. Non-Vegetarian Combo > 500 Rs.")
    print("3. Vegetarian & Non-Vegetarian Combo > 750 Rs.")
    print("4. Exit")

    choice_dish = int(input("Enter your choice: "))

    if choice_dish == 4:
        print("Exiting the process. No restaurant bill calculated.")
        return

    quantity = int(input("Enter Quantity: "))

```



```

if choice_dish == 1:
    restaurantBill = quantity * 300
elif choice_dish == 2:
    restaurantBill = quantity * 500
elif choice_dish == 3:
    restaurantBill = quantity * 750
else:
    print("Invalid choice! Exiting the process.")
    return

```

```

sql = "INSERT INTO RESTAURANT (CID, CUISINE, QUANTITY, BILL) VALUES (?, ?, ?, ?)"
values = (cid, choice_dish, quantity, restaurantBill)
cursor.execute(sql, values)
connection.commit()
print("Your total bill amount is: Rs.", restaurantBill)
print("\n\n**** WE HOPE YOU WILL ENJOY YOUR MEAL ****\n\n")
cursor.close()
else:
    print("\nError establishing SQLite connection!")

```

GAMING MODULE

```

def gaming():
    global cid, gamingBill, new_customer_added

    if new_customer_added:
        print("\nNew customer added, skipping Customer ID prompt.")
    else:
        print("Enter customer details for gaming bill.")
        customer_detail = input("Enter Customer ID or Name: ")

        # Use the entered detail to search for the customer
        cursor = connection.cursor()
        cursor.execute("SELECT * FROM C_DETAILS WHERE CID = ? OR C_NAME = ?",
(customer_detail, customer_detail))
        customer = cursor.fetchone()

        if customer:
            cid = customer[0] # Accessing by index instead of key
        else:
            print("Customer not found!")
            return

    if connection:
        cursor = connection.cursor()

```

```
cursor.execute('''
    CREATE TABLE IF NOT EXISTS GAMING (
        CID TEXT,
        GAME_NAME TEXT,
        GAME_TIME INT,
        BILL INT,
        FOREIGN KEY(CID) REFERENCES C_DETAILS(CID)
    )
''')
```

```
print("1. Virtual Reality > 1500 Rs.")
print("2. Arcade > 1000 Rs.")
print("3. Racing > 2000 Rs.")
print("4. Escape Room > 2500 Rs.")
print("5. Exit")
```

```
choice_game = int(input("Enter your choice: "))
```

```
if choice_game == 5:
    print("Exiting the process. No gaming bill calculated.")
    return
```

```
game_time = int(input("Enter Game Time (in hours): "))
```

```
if choice_game == 1:
    gamingBill = game_time * 1500
elif choice_game == 2:
    gamingBill = game_time * 1000
elif choice_game == 3:
    gamingBill = game_time * 2000
elif choice_game == 4:
    gamingBill = game_time * 2500
else:
    print("Invalid choice! Exiting the process.")
    return
```

```
sql = "INSERT INTO GAMING (CID, GAME_NAME, GAME_TIME, BILL) VALUES (?, ?, ?, ?)"
```

```
values = (cid, choice_game, game_time, gamingBill)
cursor.execute(sql, values)
connection.commit()
print("Your total gaming bill is: Rs.", gamingBill)
cursor.close()
```

```
else:
    print("\nError establishing SQLite connection!")
```

```

# FASHION MODULE
def fashionBill():
    global cid, fashionBill, new_customer_added

    if new_customer_added:
        print("\nNew customer added, skipping Customer ID prompt.")
    else:
        print("Enter customer details for fashion bill.")
        customer_detail = input("Enter Customer ID or Name: ")

        # Use the entered detail to search for the customer
        cursor = connection.cursor()
        cursor.execute("SELECT * FROM C_DETAILS WHERE CID = ? OR C_NAME = ?",
(customer_detail, customer_detail))
        customer = cursor.fetchone()

        if customer:
            cid = customer[0] # Accessing by index instead of key
        else:
            print("Customer not found!")
            return

    if connection:
        cursor = connection.cursor()
        cursor.execute('''
            CREATE TABLE IF NOT EXISTS FASHION (
                CID TEXT,
                ITEM_NAME TEXT,
                QUANTITY INT,
                BILL INT,
                FOREIGN KEY(CID) REFERENCES C_DETAILS(CID)
            )
        ''')

        print("1. T-Shirt > 500 Rs.")
        print("2. Jeans > 1000 Rs.")
        print("3. Jacket > 1500 Rs.")
        print("4. Shoes > 2000 Rs.")
        print("5. Exit")

        choice_item = int(input("Enter your choice: "))

        if choice_item == 5:
            print("Exiting the process. No fashion bill calculated.")

```

```
return
```

```
quantity = int(input("Enter Quantity: "))
```

```
if choice_item == 1:
```

```
    fashionBill = quantity * 500
```

```
elif choice_item == 2:
```

```
    fashionBill = quantity * 1000
```

```
elif choice_item == 3:
```

```
    fashionBill = quantity * 1500
```

```
elif choice_item == 4:
```

```
    fashionBill = quantity * 2000
```

```
else:
```

```
    print("Invalid choice! Exiting the process.")
```

```
    return
```

```
sql = "INSERT INTO FASHION (CID, ITEM_NAME, QUANTITY, BILL) VALUES (?, ?,  
?, ?)"
```

```
values = (cid, choice_item, quantity, fashionBill)
```

```
cursor.execute(sql, values)
```

```
connection.commit()
```

```
print("Your total fashion bill is: Rs.", fashionBill)
```

```
cursor.close()
```

```
else:
```

```
    print("\nError establishing SQLite connection!")
```

```
# FINAL BILL MODULE
```

```
def showTotalBill():
```

```
    global roomRentAmount, restaurantBill, gamingBill, fashionBill, totalAmount, cid
```

```
    print("Enter customer details to show total bill.")
```

```
    customer_detail = input("Enter Customer ID or Name: ")
```

```
    if connection:
```

```
        cursor = connection.cursor()
```

```
        # Check if the customer exists
```

```
        cursor.execute("SELECT * FROM C_DETAILS WHERE CID = ? OR C_NAME = ?",  
(customer_detail, customer_detail))
```

```
        customer = cursor.fetchone()
```

```
        if customer:
```

```
            cid = customer[0] # Accessing by index instead of key
```

```
        else:
```

```
            print("Customer not found!")
```

return

Check if a room rent record exists

```
cursor.execute("SELECT ROOM_RENT FROM ROOM_RENT WHERE CID = ?", (cid,))
```

```
result = cursor.fetchone()
```

```
if result:
```

```
    roomRentAmount = result[0]
```

```
else:
```

```
    print(f"No room rent record found for CID {cid}")
```

```
    roomRentAmount = 0
```

Check if a restaurant bill record exists

```
cursor.execute("SELECT BILL FROM RESTAURANT WHERE CID = ?", (cid,))
```

```
result = cursor.fetchone()
```

```
if result:
```

```
    restaurantBill = result[0]
```

```
else:
```

```
    print(f"No restaurant bill record found for CID {cid}")
```

```
    restaurantBill = 0
```

Check if a gaming bill record exists

```
cursor.execute("SELECT BILL FROM GAMING WHERE CID = ?", (cid,))
```

```
result = cursor.fetchone()
```

```
if result:
```

```
    gamingBill = result[0]
```

```
else:
```

```
    print(f"No gaming bill record found for CID {cid}")
```

```
    gamingBill = 0
```

Check if a fashion bill record exists

```
cursor.execute("SELECT BILL FROM FASHION WHERE CID = ?", (cid,))
```

```
result = cursor.fetchone()
```

```
if result:
```

```
    fashionBill = result[0]
```

```
else:
```

```
    print(f"No fashion bill record found for CID {cid}")
```

```
    fashionBill = 0
```

Calculate total amount

```
totalAmount = roomRentAmount + restaurantBill + gamingBill + fashionBill
```

Show the final bill

```
print("\n\n***** FINAL BILL *****")
```

```
print("Customer ID:", cid)
```

```
print("=====")
```

```

print("Room Rent: Rs.", roomRentAmount)
print("Restaurant Bill: Rs.", restaurantBill)
print("Gaming Bill: Rs.", gamingBill)
print("Fashion Bill: Rs.", fashionBill)
print("=====")
print("Total Amount: Rs.", totalAmount)
print("*****\n")
else:
    print("\nError establishing SQLite connection!")

# Search customer
def searchCustomer():
    if connection:
        cursor = connection.cursor()
        customer_detail = input("Enter Customer ID or Name: ")
        cursor.execute("SELECT * FROM C_DETAILS WHERE CID = ? OR C_NAME = ?",
(customer_detail, customer_detail))
        customer = cursor.fetchone()

        if customer:
            print("\nCustomer found:")
            print(f"ID: {customer[0]}")
            print(f"Name: {customer[1]}")
            print(f"Address: {customer[2]}")
            print(f"Age: {customer[3]}")
            print(f"Country: {customer[4]}")
            print(f"Contact Number: {customer[5]}")
            print(f>Email: {customer[6]}")
            cursor.close()
            return True
        else:
            print("Customer not found!")
            cursor.close()
            return False
    else:
        print("\nError establishing SQLite connection!")
        return False

# Main Menu
def main():
    SQLiteconnectionCheck() # Checking the SQLite connection
    while True:
        print("""
1. Add New Customer
2. Make a Booking

```

3. Room Rent
4. Restaurant Bill
5. Gaming Bill
6. Fashion Bill
7. Show Total Bill
8. Search Customer
9. Exit

""")

```
choice = int(input("\nEnter your choice: "))
```

```
if choice == 1:
```

```
    userEntry()
```

```
elif choice == 2:
```

```
    bookingRecord()
```

```
elif choice == 3:
```

```
    roomRent()
```

```
elif choice == 4:
```

```
    restaurant()
```

```
elif choice == 5:
```

```
    gaming()
```

```
elif choice == 6:
```

```
    fashionBill()
```

```
elif choice == 7:
```

```
    showTotalBill()
```

```
elif choice == 8:
```

```
    searchCustomer()
```

```
elif choice == 9:
```

```
    print("\nThank you for using the system!")
```

```
    break
```

```
else:
```

```
    print("\nInvalid choice! Please try again.")
```

```
if __name__ == "__main__":
```

```
    main()
```

Output Screens

Main Screen

```
*IDLE Shell 3.13.0*
File Edit Shell Debug Options Window Help
Python 3.13.0 (tags/v3.13.0:60403a5, Oct 7 2024, 09:38:07) [MSC v.1941 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: K:\@Ak\@temp\IP temp\Project Work IP\Main.Project.IP.XII.2025.aks242424\hotel_management_system.py

CONGRATULATIONS! Your SQLite connection has been established!

1. Add New Customer
2. Make a Booking
3. Room Rent
4. Restaurant Bill
5. Gaming Bill
6. Fashion Bill
7. Show Total Bill
8. Search Customer
9. Exit

Enter your choice: |
```

CUSTOMER DETAILS

```
*IDLE Shell 3.13.0*
File Edit Shell Debug Options Window Help
Python 3.13.0 (tags/v3.13.0:60403a5, Oct 7 2024, 09:38:07) [MSC v.1941 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: K:\@Ak\@temp\IP temp\Project Work IP\Main.Project.IP.XII.2025.aks242424\hotel_management_system.py

CONGRATULATIONS! Your SQLite connection has been established!

1. Add New Customer
2. Make a Booking
3. Room Rent
4. Restaurant Bill
5. Gaming Bill
6. Fashion Bill
7. Show Total Bill
8. Search Customer
9. Exit

Enter your choice: 1

1. Add New Customer
9. Exit

Enter your choice: 1
Enter Customer Identification Number: C101
Enter Customer Name: Harry Potter
Enter Customer Address: New York
Enter Customer Age: 29
Enter Customer Country: America
Enter Customer Contact Number: 212-456-7890
Enter Customer Email: harry_potter90@gmail.com

New Customer entered in the system successfully!
```


CUSTOMER BOOKING RECORDS

```
*IDLE Shell 3.13.0*
File Edit Shell Debug Options Window Help
Python 3.13.0 (tags/v3.13.0:60403a5, Oct 7 2024, 09:38:07) [MSC v.1941 64 bit (AMD64)]
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: K:\@Ak\@temp\IP temp\Project Work IP\Main.Project.IP.XII.2025.aks242424\h
CONGRATULATIONS! Your SQLite connection has been established!

1. Add New Customer
2. Make a Booking
3. Room Rent
4. Restaurant Bill
5. Gaming Bill
6. Fashion Bill
7. Show Total Bill
8. Search Customer
9. Exit

Enter your choice: 2
Enter customer details to make a booking.
Enter Customer ID or Name: C101

#### We have the following rooms for you ####
1. Ultra Royal > 10000 Rs. per day
2. Royal > 5000 Rs. per day
3. Elite > 3500 Rs. per day
4. Budget > 2500 Rs. per day
5. Exit
Enter your choice: 4
Enter Start Date (YYYY/MM/DD): 2024/11/12
Enter End Date (YYYY/MM/DD): 2024/11/17
Thank you, your room has been booked from 2024/11/12 to 2024/11/17 for 5 days.
Your total booking price is: Rs. 12500
```

ROOM RENT

```
*IDLE Shell 3.13.0*
File Edit Shell Debug Options Window Help
9. Exit

Enter your choice: 3
Enter customer details to calculate room rent.
Enter Customer ID or Name: 1
Customer not found!

1. Add New Customer
2. Make a Booking
3. Room Rent
4. Restaurant Bill
5. Gaming Bill
6. Fashion Bill
7. Show Total Bill
8. Search Customer
9. Exit

Enter your choice: 3
Enter customer details to calculate room rent.
Enter Customer ID or Name: C101

#### We have the following rooms for you ####
1. Ultra Royal > 10000 Rs.
2. Royal > 5000 Rs.
3. Elite > 3500 Rs.
4. Budget > 2500 Rs.
5. Exit
Enter your choice: 3
Enter Customer Room No: 201
Enter No. of Days: 3
Thank you, your room has been booked for: 3 days
Your total room rent is: Rs. 10500
```

RESTAURENT BILL

```
Enter your choice: 4
Enter customer details for restaurant bill.
Enter Customer ID or Name: C101
1. Vegetarian Combo > 300 Rs.
2. Non-Vegetarian Combo > 500 Rs.
3. Vegetarian & Non-Vegetarian Combo > 750 Rs.
4. Exit
Enter your choice: 2
Enter Quantity: 2
Your total bill amount is: Rs. 1000
```

**** WE HOPE YOU WILL ENJOY YOUR MEAL ****

GAMING BILL

```
Enter your choice: 5
Enter customer details for gaming bill.
Enter Customer ID or Name: C101
1. Virtual Reality > 1500 Rs.
2. Arcade > 1000 Rs.
3. Racing > 2000 Rs.
4. Escape Room > 2500 Rs.
5. Exit
Enter your choice: 1
Enter Game Time (in hours): 2
Your total gaming bill is: Rs. 3000
```

FASHION STORE BILL

```
Enter your choice: 6
Enter customer details for fashion bill.
Enter Customer ID or Name: C101
1. T-Shirt > 500 Rs.
2. Jeans > 1000 Rs.
3. Jacket > 1500 Rs.
4. Shoes > 2000 Rs.
5. Exit
Enter your choice: 1
Enter Quantity: 1
Your total fashion bill is: Rs. 500
```

TOTAL BILL

```
Enter your choice: 7
Enter customer details to show total bill.
Enter Customer ID or Name: C101
```

```
***** FINAL BILL *****
Customer ID: C101
=====
Room Rent: Rs. 10500
Restaurant Bill: Rs. 1000
Gaming Bill: Rs. 3000
Fashion Bill: Rs. 500
=====
Total Amount: Rs. 15000
*****
```

1. Add New Customer
2. Make a Booking
3. Room Rent
4. Restaurant Bill
5. Gaming Bill
6. Fashion Bill
7. Show Total Bill
8. Search Customer
9. Exit

```
Enter your choice: |
```

DETAILS OF THE CUSTOMER

```
Enter your choice: 8
Enter Customer ID or Name: C101
```

```
Customer found:
ID: C101
Name: Harry Potter
Address: New York
Age: 29
Country: America
Contact Number: 212-456-7890
Email: harry_potter90@gmail.com
```

1. Add New Customer
2. Make a Booking
3. Room Rent
4. Restaurant Bill
5. Gaming Bill
6. Fashion Bill
7. Show Total Bill
8. Search Customer
9. Exit

```
Enter your choice: |
```

EXIT

```
Enter your choice: 8
Enter Customer ID or Name: C101
```

```
Customer found:
ID: C101
Name: Harry Potter
Address: New York
Age: 29
Country: America
Contact Number: 212-456-7890
Email: harry_potter90@gmail.com
```

1. Add New Customer
2. Make a Booking
3. Room Rent
4. Restaurant Bill
5. Gaming Bill
6. Fashion Bill
7. Show Total Bill
8. Search Customer
9. Exit

```
Enter your choice: 9
```

```
Thank you for using the system!
```

```
>>> |
```

MYSQL DATABASE AND TABLES USED IN THIS PROJECT

DATABASE

	name
1	C_DETAILS
2	ROOM_BOOKINGS
3	ROOM_RENT
4	RESTAURANT
5	GAMING
6	FASHION

C_DETAILS TABLE STRUCTURE

Table: C_DETAILS		
Column	Data type	Constraints
CID	TEXT	•PRIMARY KEY
C_NAME	TEXT	
C_ADDRESS	TEXT	
C_AGE	INT	
C_COUNTR Y	TEXT	
P_NO	INT	
C_EMAIL	TEXT	

ROOM_BOOKINGS TABLE STRUCTURE

Table: ROOM_BOOKINGS		
Column	Data type	Constraints
CID	TEXT	
ROOM_CHOICE	INT	
START_DATE	TEXT	
END_DATE	TEXT	
TOTAL_PRICE	INT	
Global table constraints		
•FOREIGN KEY(CID) REFERENCES C_DETAILS(CID)		

ROOM_RENT TABLE STRUCTURE

Table: ROOM_RENT		
Column	Data type	Constraints
CID	TEXT	
ROOM_CHOICE	INT	
NO_OF_DAYS	INT	
ROOM_NO	INT	
ROOM_RENT	INT	
Global table constraints		
•FOREIGN KEY(CID) REFERENCES C_DETAILS(CID)		

RESTAURANT TABLE STRUCTURE

Table: RESTAURANT		
Column	Data type	Constraints
CID	TEXT	
CUISINE	TEXT	
QUANTITY	INT	
BILL	INT	
Global table constraints		
•FOREIGN KEY(CID) REFERENCES C_DETAILS(CID)		

GAMING TABLE STRUCTURE

Table: GAMING		
Column	Data type	Constraints
CID	TEXT	
GAME_NAME	TEXT	
GAME_TIME	INT	
BILL	INT	
Global table constraints		
•FOREIGN KEY(CID) REFERENCES C_DETAILS(CID)		

FASHION TABLE STRUCTURE

Table: FASHION		
Column	Data type	Constraints
CID	TEXT	
ITEM_NAME	TEXT	
QUANTITY	INT	
BILL	INT	
Global table constraints		
•FOREIGN KEY(CID) REFERENCES C_DETAILS(CID)		

BACKEND DATA GENERATED THROUGH SOFTWARE

Table: C_DETAILS					
#	CID TEXT	C_NAME TEXT	C_ADDRESS TEXT	C_AGE INT	C_COUNTRY TEXT
0	C101	Harry Potter	New York	28	America
1	C103	Naruto Uzumaki	Konoha	19	Hidden Leaf Village
2	C105	Monkey D Luffy	Foosha Village	19	Dawn Island
3	C106	Tsunade Senju	Konoha	55	Land of Fire

BACKEND DATA GENERATED THROUGH SOFTWARE

Table: ROOM_BOOKINGS					
#	CID TEXT	ROOM_CHOICE INT	START_DATE TEXT	END_DATE TEXT	TOTAL_PRICE INT
0	C101	2	2024/10/12	2024/10/24	60000
1	C105	3	2024/10/15	2024/10/20	21000
2	C101	4	2024/11/01	2024/11/10	25000
3	C106	1	2024/12/05	2024/11/13	90000

BACKEND DATA GENERATED THROUGH SOFTWARE

Table: ROOM_RENT					
#	CID TEXT	ROOM_CHOICE INT	NO_OF_DAYS INT	ROOM_NO INT	ROOM_RENT INT
0	C101	1	2	201	20000
1	C103	4	4	205	10000
2	C106	2	1	301	5000
3	C105	3	5	105	17500

BACKEND DATA GENERATED THROUGH SOFTWARE

Table: RESTAURANT				
#	CID TEXT	CUISINE TEXT	QUANTITY INT	BILL INT
0	C101	2	2	1000
1	C103	1	1	300
2	C105	1	2	600
3	C106	3	3	2250

BACKEND DATA GENERATED THROUGH SOFTWARE

Table: GAMING				
#	CID TEXT	GAME_NAM E TEXT	GAME_TIM E INT	BILL INT
0	C101	1	2	3000
1	C103	4	1	2500
2	C105	3	2	4000
3	C106	2	1	1000

BACKEND DATA GENERATED THROUGH SOFTWARE

Table: FASHION				
#	CID TEXT	ITEM_NAM E TEXT	QUANTIT Y INT	BILL INT
0	C101	4	2	4000
1	C103	3	1	1500
2	C105	1	3	1500
3	C106	2	2	2000

Hardware and Software Requirements

1. Hardware Requirement

- Pentium 3/4/core 2 Duo/Dual core/i3/i5/i7
- With at least 256 MB RAM
- 5MB free space on Hard Disk
- Color Monitor/LCD

2. Software Requirement

- MS Windows
- Python with related library used for data analysis
 - Pandas
 - Numpy
 - Matplotlib
- SQ lite for RDBMS.